

Echo-Corrected Information Diffusion: Certified Multiscale Phase-Type Hawkes Processes for Macro-News Absorption across FX, Equities, Gold and Crypto

Sayak Dutta

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Abstract

How fast do prices absorb scheduled macroeconomic news, and how much of the post-release activity is the market reacting to itself? We model δ -crossing price events of six markets—EUR/USD, USD/JPY, gold, the S&P 500, bitcoin and ether—around 670 scheduled releases (2022–2026) with a multiscale phase-type Hawkes process whose parameters all enter linearly. The log-likelihood is then concave, and we compute the global maximum with a Fenchel duality-gap certificate. A Markov embedding gives the echo-corrected response to a release in closed form; we prove it is stable iff the spectral radius of the branching matrix is below one. A built-in placebo test on matched release-free days, with a penalty chosen on held-out windows, separates genuine news effects from artefacts: CPI, payrolls, PPI, the FOMC statement and retail sales pass it, second-tier releases and the ECB and BoJ do not. For the releases that pass, the direct reaction is absorbed in about a second, but echoes stretch half of the response to 16–137 s and ninety percent to 5–11 minutes; the branching ratio is $\rho(G) = 0.87$ and declines from 2022 to 2026. On synthetic benchmarks the certified estimator recovers multiscale and network kernels with 7–10 $\times$ smaller likelihood deficits than six classical estimators. We also introduce EPT-X, a neural point process with an exact compensator that superposes a history-conditioned hyper-Erlang renewal channel on a linear Hawkes backbone; on LOBSTER limit-order-book data it achieves the best test likelihood on all five stocks against six classical estimators and seven neural baselines, by 0.23–0.41 nats per event over the best neural model. On the public EasyTPP benchmarks it improves the best published likelihood on Retweet by more than 0.6 nats per event and matches it on Taobao and StackOverflow. Along the way we find that a widely used IntensityFree implementation clamps inter-event times before scoring, which overstates its likelihood on streams with many short gaps.

1 Introduction

Scheduled macroeconomic releases are the cleanest natural experiments in finance: the time of the shock is known to the second, its content is unknown until release, and its surprise relative to expectations can be measured. How fast prices absorb such information, and how much of the post-release activity is information versus the market reacting to itself, are old questions [2, 10]. Hawkes processes answer the second question elegantly—the branching structure separates exogenous (news-driven) events from endogenous (self-excited) ones—and Rambaldi, Pennesi and Lillo [10] showed that adding a news kernel to a Hawkes model of FX quote activity captures the post-announcement burst.

Three gaps remain. *First*, the absorption speed is usually read off the news kernel alone (the “half-life” $\ln 2/\beta$ of an exponential), which ignores the endogenous echoes triggered by the news; when the branching ratio is close to one [6, 8] the echoes dominate the total response. *Second*, estimation is fragile: flexible kernels are fitted by local optimisation of a non-convex likelihood, and branching ratios are biased upward when seasonality is mis-specified [7]. *Third*, the information content of the release (its surprise) enters, if at all, as a post-hoc correlate rather than as part of the model, and cross-market propagation is rarely modelled jointly.

Contributions. Linear-in-parameters kernel dictionaries, EM for Hawkes processes and the Markov embedding of Erlang-kernel Hawkes processes are known individually; our contribution is to combine them into a measurement framework whose output—echo-corrected absorption times—is certified and falsifiable.

1. **A certified estimator.** We represent every kernel—endogenous, news, anticipatory—in a phase-type dictionary (exponential and Erlang phases on a log grid of time scales). All parameters enter the intensity linearly, the log-likelihood is concave, and we compute the global maximum with accelerated EM and a barrier-Newton polish, returning a Fenchel duality gap that certifies optimality (Proposition 2).
2. **Echo-corrected absorption in closed form.** A Markov embedding turns the mean response to a release into $Ce^{At}\mathbf{z}_0$. We prove that the embedding is stable iff the spectral radius of the branching matrix is below one (Theorem 3) and obtain absorption quantiles, echo amplification and—via δ -crossing sampling—exact expected price-discovery curves (Corollary 4).
3. **Identification by design.** Per-window baselines, time-of-day hats identified by matched placebo windows, and a PLACEBO news type whose fitted kernel must vanish give a falsification test built into the likelihood.
4. **Surprise as a mark.** Standardised surprises from real-time nowcasts (Cleveland Fed CPI nowcast, Atlanta Fed GDPNow) and real-time statistical expectations enter the news kernel linearly and separately by sign.
5. **EPT-TTP.** A neural temporal point process whose latent state is a nonnegative phase-type system: exact compensator, Bayes-optimal predictions, and it nests both multivariate Hawkes processes and RMTTP (Section 4). Its extension EPT-X superposes a history-conditioned hyper-Erlang renewal channel, which keeps the compensator exact while making the conditional gap law universal (Section 4.1).
6. **Benchmarks.** Synthetic recovery against six classical estimators, public neural-TTP benchmarks under the S2P2 protocol, LOBSTER limit-order-book streams, and post-release forecasting on a 2022–2026 six-asset news panel.

2 The MSX-Hawkes model

Events. For each asset a we record δ -crossing events: an up (down) event whenever the log mid-price moves $+\delta_a$ ($-\delta_a$) away from the level of the previous event, with a jump of $n\delta_a$ producing n tied events. This intrinsic-time sampling filters bid–ask bounce, makes assets comparable in basis points and, crucially, turns intensities into prices: the log-price change over any interval equals $\delta_a(N_a^\uparrow - N_a^\downarrow)$ up to one δ_a . Dimension $2a$ ($2a + 1$) counts up (down) crossings of asset a ; there are $d = 2A$ dimensions.

Phase-type dictionary. Fix rates $\beta_1, \dots, \beta_K$ on a logarithmic grid and Erlang orders $r = 0, \dots, R-1$. The normalised basis densities

$$g_{kr}(t) = \beta_k \frac{(\beta_k t)^r}{r!} e^{-\beta_k t}, \quad \int_0^\infty g_{kr} = 1, \quad (1)$$

generate kernels $\phi(t) = \sum_{k,r} a_{kr} g_{kr}(t)$ with $a_{kr} \geq 0$ and branching ratio $\sum_{k,r} a_{kr}$. Exponential mixtures ($R=1$) approximate every completely monotone kernel (Bernstein's theorem), including power laws; higher orders add humps ($\phi(0)=0$), and phase-type densities are dense in the densities on $\mathbb{R}_+$.

Intensity. For target dimension i , window w with likelihood interval $[t_0^w, t_1^w)$, releases e at times τ_e of type c_e with nonnegative mark features $\mathbf{z}_e = (1, z_e^+, z_e^-)$ (standardised surprise split by sign),

$$\begin{aligned} \lambda_i(t) = & \underbrace{c_{iw} + \sum_m \eta_{im} h_m(\text{tod}(t))}_{\text{baseline}} + \underbrace{\sum_j \sum_{t_n^j < t} \sum_{k,r} a_{ijk} g_{kr}(t - t_n^j)}_{\text{endogenous}} \\ & + \underbrace{\sum_{e:\tau_e < t} \sum_{k,r,m} b_{ic_ekrm} z_{em} g_{kr}^{\text{exo}}(t - \tau_e)}_{\text{news}} + \underbrace{\sum_{e:\tau_e > t} \sum_k \alpha_{ic_ek} \beta_k^{\text{ant}} e^{-\beta_k^{\text{ant}}(\tau_e - t)}}_{\text{anticipation}}, \end{aligned} \quad (2)$$

with per-window intercepts c_{iw} , periodic hat functions h_m of local time of day (identified by matched placebo windows, Section 5), and all parameters nonnegative. Writing $\mathbf{x}(t) \geq 0$ for the stacked features, $\lambda_i(t) = \mathbf{x}(t)^\top \boldsymbol{\theta}_i$.

Likelihood. The log-likelihood separates across targets,

$$\ell_i(\boldsymbol{\theta}_i) = \sum_{n: u_n = i} \log(\mathbf{x}(t_n)^\top \boldsymbol{\theta}_i) - \boldsymbol{\Phi}^\top \boldsymbol{\theta}_i, \quad \boldsymbol{\Phi} = \sum_w \int_{t_0^w}^{t_1^w} \mathbf{x}(t) dt, \quad (3)$$

and every entry of $\boldsymbol{\Phi}$ and of $\mathbf{x}(t_n)$ is available in closed form: the endogenous features follow the exact Erlang-chain recursion of Lemma 1 in $O(NdKR^2)$ time, and integrals are regularised incomplete gamma functions.

3 Theory

3.1 Exact recursion

Lemma 1 (Erlang-chain recursion). *Let $s_r(t) = \sum_{t_n < t} g_{kr}(t - t_n)$ for one source and one rate β . Between events, $\mathbf{s} = (s_0, \dots, s_{R-1})$ solves $\dot{\mathbf{s}} = \beta(N - I)\mathbf{s}$ with N the down-shift, and at an event $s_0 += \beta$. Hence, with $y = \beta\Delta$ and $w_j = e^{-y}y^j/j!$,*

$$s_r(t + \Delta) = \sum_{j \leq r} w_j s_{r-j}(t), \quad \int_t^{t+\Delta} s_r = \frac{1}{\beta} \sum_{j \leq r} P(j+1, y) s_{r-j}(t),$$

where P is the regularised lower incomplete gamma function.

Proof. $\frac{d}{dt}g_{kr} = \beta(g_{k,r-1} - g_{kr})$ for $r \geq 1$ and $\frac{d}{dt}g_{k0} = -\beta g_{k0}$, with $g_{kr}(0) = \beta \mathbb{K}[r=0]$. The matrix exponential of $\beta(N - I)\Delta$ is $e^{-y} \sum_{j < R} \frac{y^j}{j!} N^j$ since N is nilpotent and commutes with I ; integrating term by term gives $P(j+1, y)/\beta$. $\square$

3.2 Concavity, EM and a certificate of global optimality

Problem (3) is the maximisation of a concave function over the nonnegative orthant (a Poisson linear-inverse problem). Nonnegative ℓ_1 penalties are linear, $\gamma^\top \theta$, and are absorbed into Φ . The multiplicative EM map

$$\theta_p \leftarrow \theta_p \frac{\sum_n x_{np}/\lambda_n}{\Phi_p + \gamma_p}$$

is monotone and keeps iterates positive; we accelerate it with SQUAREM and polish with a log-barrier Newton method whose Hessian $\sum_n \mathbf{x}_n \mathbf{x}_n^\top / \lambda_n^2$ is assembled blockwise (dense endogenous block, sparse baseline/news block).

Proposition 2 (Duality-gap certificate). *For any $\mathbf{v} > 0$ with $X^\top \mathbf{v} \leq \Phi + \gamma$ and any feasible θ ,*

$$\ell(\theta) \leq D(\mathbf{v}) := -N - \sum_n \log v_n.$$

Taking $\mathbf{v} = \lambda^{-1}/s$ with $s = \max_p (X^\top \lambda^{-1})_p / (\Phi_p + \gamma_p)$ gives the computable bound $\ell^* - \ell(\theta) \leq D(\mathbf{v}) - \ell(\theta) = N \log s + (\theta^\top (\Phi + \gamma) - N)$.

Proof. Fenchel's inequality $\log u \leq vu - 1 - \log v$ ($u, v > 0$) applied to $u = \lambda_n$ gives $\sum_n \log \lambda_n \leq \theta^\top X^\top \mathbf{v} - N - \sum_n \log v_n$, and $\theta^\top X^\top \mathbf{v} \leq \theta^\top (\Phi + \gamma)$ for $\theta \geq 0$; subtracting $\theta^\top (\Phi + \gamma)$ from both sides yields $\ell(\theta) \leq D(\mathbf{v})$ for every feasible θ , hence also for the maximiser. With $v_n = 1/(s\lambda_n)$, $-\sum_n \log v_n = \sum_n \log \lambda_n + N \log s$, which gives the stated gap. Both terms vanish at a KKT point, where $s = 1$ and $\theta^\top (\Phi + \gamma) = \sum_n \theta^\top \mathbf{x}_n / \lambda_n = N$. $\square$

At the optimum $s = 1$ (KKT), and the expected number of events $\theta^\top \Phi$ equals the observed number N_i for unpenalised fits—both are checked in our tests.

3.3 Markov embedding and echo-corrected responses

Stack the states s_{jkr} of all sources into $\mathbf{x}$, let F be the block-diagonal Erlang-chain generator, G_{in} inject β_k into phase $r = 0$ of source j , and Θ the endogenous weights, so $\lambda = \mu + \Theta \mathbf{x}$. Taking expectations of the jump dynamics, the *mean* excess intensity caused by one release with news states $\mathbf{y}$ ($\dot{\mathbf{y}} = F_e \mathbf{y}$, $\mathbf{y}(0) = \mathbf{y}_0$) satisfies

$$\dot{\mathbf{z}} = \mathcal{A} \mathbf{z}, \quad \mathcal{A} = \begin{pmatrix} F + G_{\text{in}} \Theta & G_{\text{in}} \Theta_e \\ 0 & F_e \end{pmatrix}, \quad r(t) = \begin{pmatrix} \Theta & \Theta_e \end{pmatrix} \mathbf{z}(t), \quad \mathbf{z}(0) = \begin{pmatrix} 0 \\ \mathbf{y}_0 \end{pmatrix}. \quad (4)$$

Theorem 3 (Stability). *Let $G_{ij} = \sum_{k,r} a_{ijk r}$ be the branching matrix. Then $F + G_{\text{in}} \Theta$ is Hurwitz if and only if $\rho(G) < 1$.*

Proof. F is Metzler and Hurwitz ($-F$ is a nonsingular M-matrix: lower-triangular blocks with diagonal $-\beta_k < 0$) and $G_{\text{in}} \Theta \geq 0$. For a Metzler matrix $F + P$ with F Hurwitz-Metzler and $P \geq 0$, $F + P$ is Hurwitz iff $\rho(-F^{-1}P) < 1$ (regular splitting of the M-matrix $-(F + P)$; Varga, Thm. 3.29). Now $\rho(-F^{-1}G_{\text{in}}\Theta) = \rho(\Theta(-F)^{-1}G_{\text{in}})$ and $(\Theta(-F)^{-1}G_{\text{in}})_{ij} = \sum_{k,r} a_{ijk r} \int_0^\infty g_{kr} = G_{ij}$, because $(-F)^{-1}G_{\text{in}}\mathbf{e}_j$ collects the integrals of the basis densities excited by one event of source j . $\square$

Corollary 4 (Closed-form absorption). *Under $\rho(G) < 1$, the echo-corrected response is $r(t) = C e^{\mathcal{A}t} \mathbf{z}_0$, its total is $\int_0^\infty r = -C \mathcal{A}^{-1} \mathbf{z}_0$, the cumulative fraction absorbed by time t is*

$$F(t) = \frac{C \mathcal{A}^{-1} (e^{\mathcal{A}t} - I) \mathbf{z}_0}{C \mathcal{A}^{-1} (-\mathbf{z}_0)},$$

the echo-corrected absorption quantile is $t_q = F^{-1}(q)$, and the amplification (echo multiplier) is $\int r / \int r_{\text{direct}}$ with $r_{\text{direct}}(t) = \Theta_e e^{F_e t} \mathbf{y}_0$. For an asset with dimensions (u, v) the expected price-discovery curve is $\delta_a \int_0^t (r_u - r_v)$.

Remark 5. The popular “half-life $\ln 2/\beta$ ” of a single exponential news kernel measures only r_{direct} ; when $\rho(G)$ is close to one the echoes dominate and t_{50} of the full response can be orders of magnitude longer. Section 7 quantifies the gap.

3.4 Closed-form conditional forecasts

Given the pre-release history (states $\mathbf{x}(\tau)$ computed exactly by Lemma 1), the observed surprise of the release at τ , and a constant baseline $\boldsymbol{\mu}$ over the horizon, $\mathbb{E}[\mathbf{N}(\tau, \tau + h) \mid \mathcal{F}_\tau] = \boldsymbol{\mu}h + C[\mathcal{A}^{-1}(e^{Ah} - I)\mathbf{z}_0 + \mathcal{A}^{-1}(\mathcal{A}^{-1}(e^{Ah} - I) - hI)\mathbf{u}]$ with $\mathbf{u} = (G_{\text{in}}\boldsymbol{\mu}, 0)$. Under δ -sampling this is also a realised-variance ($\delta_a^2 N$) and drift forecast (Track D).

4 EPT-TPP: a neural point process with an exact compensator

Neural marked TPPs (RMTPP, NHP, THP, SAHP, AttNHP, S2P2) either restrict the intensity to a simple parametric form between events or estimate the compensator $\int \lambda$ by Monte Carlo. We keep the phase-type algebra of Section 3 and let a recurrent network drive it.

State and dynamics. The latent state $\mathbf{x}(t) \in \mathbb{R}_{\geq 0}^P$ consists of C channels of Erlang- R chains with learnable rates β_{ck} . Between events it evolves as $\dot{\mathbf{x}} = \mathbf{F}_\beta \mathbf{x}$ (Lemma 1). At event n with mark m_n and gap Δ_n , a GRU summary $\mathbf{h}_n$ produces a gate and a nonnegative jump,

$$\mathbf{x}(t_n^+) = \sigma(W_g \mathbf{h}_n) \odot \mathbf{x}(t_n^-) + \text{softplus}(W_j \mathbf{h}_n).$$

Intensity. For marks $m = 1, \dots, M$,

$$\lambda_m(t) = \mu_m(\mathbf{h}_n) + \sum_p C_{mp} x_p(t) + g_{1m}(\mathbf{h}_n) e^{-a(\mathbf{h}_n)s} + g_{2m}(\mathbf{h}_n)(1 - e^{-\kappa(\mathbf{h}_n)s}), \quad s = t - t_n,$$

with $C, \mu, g_1, g_2, a, \kappa \geq 0$: bounded decaying and rising-but-saturating hazards (an unbounded Gompertz term overflows on heavy-tailed gaps).

Proposition 6. (i) $\lambda_m > 0$ everywhere. (ii) The compensator between events is exact: $\int_{t_n}^{t_n + \Delta} \lambda_m = \mu_m \Delta + \mathbf{C}_m^\top \mathbf{I}(\Delta) + g_{1m}(1 - e^{-a\Delta})/a + g_{2m}(\Delta - (1 - e^{-\kappa\Delta})/\kappa)$ with $\mathbf{I}(\Delta)$ given by Lemma 1. (iii) With $g = 0$, identity gates and jumps $\text{softplus}(W_j \mathbf{h}_n) = \mathbf{e}_{m_n} \otimes \boldsymbol{\beta}$ the model is the multivariate phase-type Hawkes process of Section 2; with $C = 0$ it is RMTPP [5].

Consequently training maximises the *exact* log-likelihood (no sampling variance in gradients), evaluation reports the exact test likelihood, and next-event predictions are Bayes-optimal functionals of the exact density: $\mathbb{E}[\Delta] = \int_0^\infty S(s) ds$ and $\hat{m} = \arg \max_m \int_0^\infty \lambda_m(s) S(s) ds$, computed by one-dimensional quadrature.

4.1 EPT-X: a renewal channel for within-cascade timing

Order-book streams contain cascades: a single marketable order produces several messages with the same timestamp, and inter-event gaps span ten orders of magnitude, from nanoseconds to seconds. A linear Hawkes intensity describes excitation *between* cascades well but has no direct handle on the gap distribution *within* them. EPT-X adds, by superposition, a second channel that models the *next gap* directly while keeping the compensator exact.

Defective hyper-Erlang channel. Fix a dictionary of J Erlang atoms (R_j, β_j) whose means R_j/β_j lie on a logarithmic grid spanning the data’s time scales, with orders $R_j \in \{1, 4, 16\}$ (coefficient of variation $R_j^{-1/2}$, so high orders give sharp modes in log-time). After event n the history $\mathbf{h}_n$ selects weights $\mathbf{w}(\mathbf{h}_n)$ on the simplex over the J atoms and a *defect* atom w_∞ , and a mark law $q_{m|j}(\mathbf{h}_n)$ for each time scale. The channel’s joint density of the next gap s and mark m and its survival function are

$$p(s, m) = \sum_{j=1}^J w_j q_{m|j} \frac{\beta_j^{R_j} s^{R_j-1} e^{-\beta_j s}}{(R_j - 1)!}, \quad S_X(s) = w_\infty + \sum_{j=1}^J w_j e^{-\beta_j s} \sum_{i < R_j} \frac{(\beta_j s)^i}{i!}.$$

EPT-X superposes this channel on EPT (including its Hawkes backbone): $\lambda_m(t) = \lambda_m^{\text{EPT}}(t) + p(s, m)/S_X(s)$.

Proposition 7. (i) *The compensator over $(t_n, t_n + \Delta]$ is $\Lambda^{\text{EPT}}(\Delta) - \log S_X(\Delta)$, in closed form.* (ii) *With $w_\infty = 1$ EPT-X is exactly EPT, which in turn nests the multivariate phase-type Hawkes process.* (iii) *Given $\mathbf{h}_n$, the renewal channel is the absorption time of a finite continuous-time Markov chain (a phase-type law), so its survival, quantiles and moments are closed-form matrix functionals.* (iv) *Since mixtures of Erlang distributions are dense in the distributions on $(0, \infty)$ [9, 12], the channel can approximate any conditional gap law as the dictionary is refined—the expressiveness of a conditional mixture density [11], with an exact intensity and compensator.*

The two channels have distinct roles: the renewal channel describes *within-cascade* timing and the mark mix of a cascade (the mark law depends on which time scale fires), while the Hawkes channel describes *cross-cascade* excitation. The GRU also receives the standardised log-gap, because a linear gap feature cannot distinguish nanosecond ties from zero.

5 Data

Releases. The calendar covers January 2022–August 2026 (670 scheduled events). US data releases (CPI, nonfarm payrolls, PPI, retail sales, GDP advance estimate, initial jobless claims) are dated by their *initial* vintage in ALFRED, which removes seasonal-factor-only releases automatically; all are released at 08:30 America/New_York and converted to UTC with daylight saving. FOMC statements (14:00 ET) and press conferences (14:30 ET) come from the Federal Reserve calendar; ECB decisions and press conferences from the ECB (13:45/14:30 CET until 21 July 2022 and 14:15/14:45 CET afterwards); Bank of Japan decisions from the BoJ, whose release time is not fixed (a latent onset, flagged).

Expectations and surprises. CPI expectations are the Cleveland Fed inflation nowcast for the reference month, taken from the last nowcast strictly before the release date; GDP expectations are the Atlanta Fed GDPNow forecast immediately before BEA’s advance estimate. For payrolls, claims, PPI and retail sales we use real-time statistical expectations from the sequence of first prints (mean of the previous 3, 1, 6 and 6 prints respectively); survey consensus is not freely available and this limitation is stated wherever surprises are used. Surprises are standardised by the expanding standard deviation of past surprises of the same type (no look-ahead) and winsorised at ± 4 .

Prices. FX (EUR/USD, USD/JPY), gold (XAU/USD) and an S&P 500 CFD come from Hist-Data millisecond bid/ask ticks (stamped in fixed EST, converted to UTC); BTC and ETH from Binance USD^M-futures aggregated trades. A clock check at a CPI release confirms activity jumps

within one second of the scheduled time in both sources. The HistData feed has a gap from February to July 2023 for all four instruments; windows without at least 90% of 10-minute blocks containing five ticks are excluded (per window, documented in the data card).

Events and windows. Prices are converted to δ -crossing events with δ calibrated per asset so that the median intraday rate is about 0.2 events per second: $\delta = 0.25$ (EUR/USD, USD/JPY), 0.7 (XAU/USD, S&P 500 CFD), 2.0 (BTC) and 3.0 (ETH) basis points; simultaneous crossings collapse into one event with a size mark. Releases closer than 60 minutes are merged into one window [first -30 min, last $+60$ min) with 30 minutes of burn-in history. Each window is matched with a placebo window at the same clock times on the same weekday one to six weeks away, with no modelled release and none of sixteen second-tier US releases within three hours; its pseudo-releases carry a per-type PLACEBO type.

6 Benchmark protocol

All result tables are regenerated from JSON files written by the scripts in `scripts/`; every method is run by us on identical data unless a table explicitly cites published numbers.

Track A: synthetic recovery. Five truths, simulated by exact cluster sampling, deliberately include kernels *outside* the MSX class: S1 exponential (the home turf of single-decay estimators), S2 a true power law (Lomax, tail exponent 1.5, as reported for FX and futures), S3 a gamma hump plus a fast exponential, S4 a sparse 4-dimensional multiscale network (10 ms and 30 s components), S5 a sparse 10-dimensional power-law network. Each scenario uses 10 windows per train and test set and five seeds. Competitors: tick’s exponential-kernel MLE (decay chosen by training likelihood), sum-of-exponentials MLE on the same decay grid as MSX, the nonparametric EM of Lewis–Mohler type, ADM4 [13], the conditional-law estimator [3] (quadrature chosen by training likelihood), and NPHC [1] (our PyTorch solver; tick 0.8 ships a stub). Metrics: relative Frobenius error of the branching matrix, spectral-radius error, relative L^1 kernel error on lags 10^{-4} – 10^4 s, edge AUC/AP, and held-out log-likelihood deficit per event relative to the true model (dLL , lower is better). MSX-auto chooses the Erlang order and an exposure-weighted ℓ_1 level on held-out training windows only.

Track B: public marked-TPP benchmarks. EasyTPP splits of Amazon, Retweet, Taxi, Taobao and StackOverflow and the protocol of S2P2 [4]: log-likelihood per scored event with the first event of each sequence as conditioning, model selection on validation likelihood, five seeds. Published numbers are quoted from Chang et al. [4]; our re-run of S2P2 validates the harness. EPT-TPP reports the *exact* likelihood; neural baselines estimate the compensator by Monte Carlo.

Track C: limit-order-book event streams. LOBSTER samples (AAPL, AMZN, GOOG, INTC, MSFT, 21 June 2012), six event types (limit orders, cancellations and market orders on each side), a chronological 60/20/20 split, sequences of 128 events. Neural baselines (RMTPP, NHP, THP, SAHP, AttNHP, IntensityFree, S2P2) run with the S2P2 authors’ configurations; on our 8 GB laptop GPU the batch size is reduced from 256 to 64 to avoid the Windows GPU watchdog. The classical Hawkes family is scored under the same protocol.

Track D: post-release forecasting. Chronological split of the news panel (train 2022–2024, test 2025–2026). Targets are the number of δ -crossings (activity, and realised variance $\delta^2 N$) and the signed drift in the 1, 5 and 15 minutes after each release. MSX forecasts are the closed form of Section 3 with the window baseline re-estimated from the pre-release data; baselines are same-type climatology, an event-study regression on pre-release activity and $|z|$, and a HAR regression on lagged activity.

7 Results

7.1 Track A: synthetic recovery

Table 1 reports the held-out likelihood deficit (means and standard deviations over five seeds); branching-matrix, spectral-radius and kernel errors are in Table 8 (Appendix). When the truth is a single exponential (S1) the correctly specified single-decay estimators (tick’s exponential MLE and ADM4) are best, as they must be. Under a true power law (S2) the sum-of-exponentials estimators—tick’s and MSX—are statistically tied (held-out deficit 0.2 vs. 0.3×10^{-3} nats/event; tick’s solver failed on one of five seeds). Whenever the kernel has structure outside a single family—a delayed hump (S3), components ten milliseconds and thirty seconds apart (S4), or a sparse ten-dimensional power-law network (S5)—MSX-auto is best on every metric: its held-out deficit is 7–10 times smaller than the best competitor’s, its branching-matrix error is roughly halved, and it recovers the causal graph perfectly (AUC = 1). The exponential-only ablation isolates the contribution of Erlang phases, and the certificate of Proposition 2 was at most 3.3×10^{-4} nats over all 100 MSX fits.

Table 1: Track A, synthetic recovery: held-out log-likelihood deficit to the true model (10^{-3} nats per event, lower is better; mean $\pm$ s.d. over five seeds). Bold: best; underlined: second best; shaded: our estimators.

	S1 exp	S2 power law	S3 hump	S4 multiscale	S5 network
<i>Classical estimators</i>					
Exp. kernel MLE	0.05 ± 0.05	26.05 ± 0.50	101.47 ± 1.81	711.28 ± 53.09	426.80 ± 368.57
Sum-of-exp. MLE	0.92 ± 0.13	0.18 ± 0.12	11.51 ± 0.46	—	—
ADM4	<u>0.05</u> ± 0.05	26.05 ± 0.50	101.47 ± 1.81	3.13 ± 0.25	26.51 ± 0.29
Hawkes EM (nonparam.)	1.02 ± 0.07	1.73 ± 0.21	2.24 ± 0.12	4.55 ± 0.29	5.23 ± 0.25
Conditional law	2.84 ± 1.03	4.91 ± 0.59	10.66 ± 2.13	114.94 ± 1.20	78.64 ± 4.79
<i>Ours</i>					
MSX ($R = 2$)	0.23 ± 0.12	0.32 ± 0.12	5.90 ± 0.36	0.54 ± 0.12	1.19 ± 0.07
MSX ($R = 4$)	0.28 ± 0.15	0.45 ± 0.10	<u>0.31</u> ± 0.15	0.70 ± 0.11	1.63 ± 0.09
MSX, exponential only	0.91 ± 0.13	<u>0.19</u> ± 0.11	11.51 ± 0.46	<u>0.32</u> ± 0.08	<u>0.82</u> ± 0.08
MSX-auto	0.24 ± 0.12	0.33 ± 0.13	0.30 ± 0.13	0.32 ± 0.08	0.74 ± 0.10

7.2 Main study: how fast do markets absorb scheduled macro news?

The pooled 2022–2026 model (732 windows—430 release and 302 placebo—with 7.3 million scored δ -crossing events, 8.4 million including burn-in history, in twelve dimensions) is certified optimal to within 0.0054 nats per dimension. The spectral radius of the branching matrix is $\rho(G) = 0.87$: roughly seven in every eight price events are endogenous echoes. Fitted year by year, $\rho(G)$ declines

from 2022 to 2026 (0.91, 0.89, 0.85, 0.86, 0.85), and the slowest echo mode relaxes in about 12 minutes.

Falsification. Figure 1 compares, for every release type, the direct news-kernel mass with that of a matched placebo at the same clock time on a day without any scheduled release (neither the 11 modelled types nor sixteen second-tier US releases). With the exposure-weighted ℓ_1 level selected on held-out windows, 5 of the 11 release types exceed their placebo by at least 50%: CPI (8.5 $\times$), payrolls (6.7 $\times$), PPI (4.0 $\times$), the FOMC statement (3.9 $\times$) and retail sales (1.9 $\times$). Jobless claims, ECB and BoJ decisions and the press conferences do not have a detectable *direct* effect beyond their placebos—an honest negative that the unregularised fit would have hidden: without the penalty, nonnegative news kernels absorb as much mass on placebo days as on release days (boundary bias), and the first specification with long news kernels failed the test outright (Appendix).

Echo-corrected absorption. For the releases that pass the placebo test, the direct news kernel is absorbed almost instantly (median t_{50} of 1.1 s), but the market’s own echoes stretch the response: half of the total extra activity arrives within 16–137 s and ninety percent within 5–11 minutes (Figure 2, Table 2). The textbook “news half-life” $\ln 2/\beta$ therefore understates absorption time by one to two orders of magnitude.

Price discovery. Signed surprises enter the news kernel separately by sign, so the model yields expected drift curves (Figure 3). The signs are economically coherent: hotter-than-expected inflation and stronger payrolls depress crypto and lift the dollar, while cooler prints do the opposite.

Goodness of fit. Time-rescaled residuals have Kolmogorov–Smirnov statistics of at most 0.044 against Exp(1), and the per-window excess-dispersion test rejects in 26–55% of windows, compared with 42% (power-law kernel) and 93% (double exponential) reported for EBS FX data by Rambaldi et al. [10].

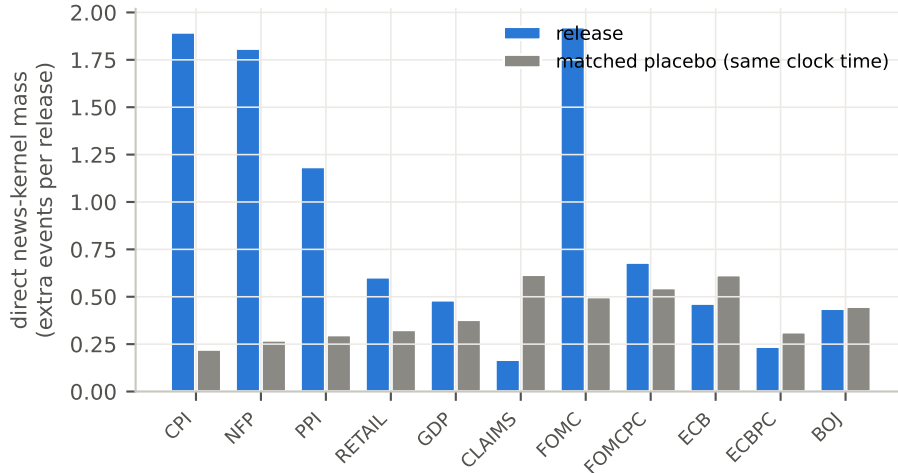


Figure 1: Placebo test. Direct news-kernel mass (expected extra events per release, all assets) for each release type and for a matched placebo at the same clock time on a release-free day.

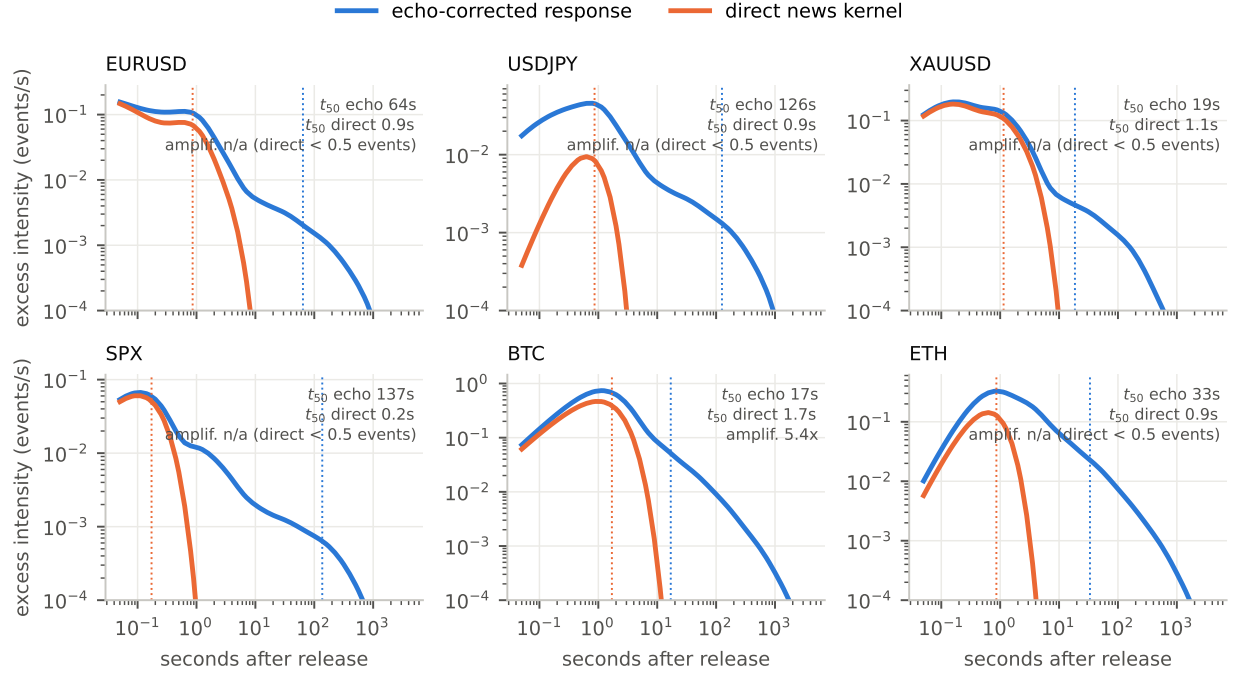


Figure 2: Echo-corrected response to a CPI release (blue) versus the direct news kernel (orange), per asset; dotted lines mark t_{50} .

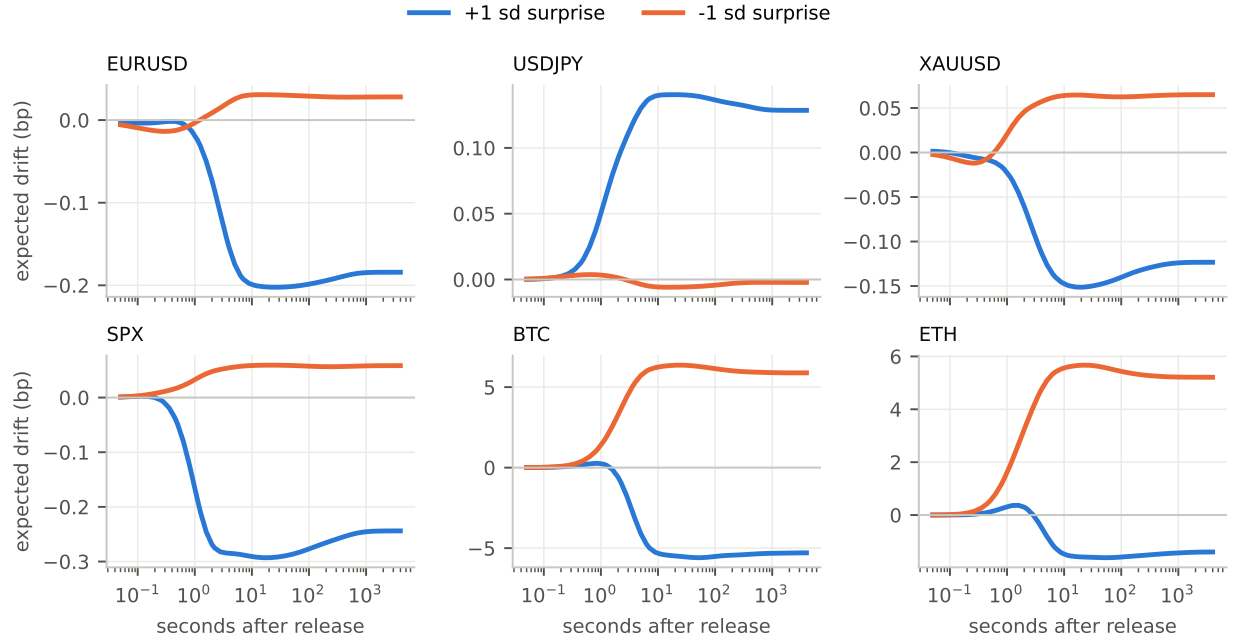


Figure 3: Expected price-discovery curves after a ± 1 standard-deviation CPI surprise.

Table 2: Echo-corrected absorption by release and asset (pooled model): t_{50} in seconds / t_{90} in minutes after the release. *Release types that do not exceed their matched placebo by 50%; their absorption times describe a statistically undetected effect.

Release	EURUSD	USDJPY	Gold	S&P 500	BTC	ETH
CPI	64 / 9.1	126 / 11.0	19 / 4.8	137 / 11.0	17 / 6.1	33 / 7.8
NFP	94 / 10.1	109 / 10.7	16 / 5.2	80 / 8.8	27 / 7.5	24 / 7.3
PPI	120 / 10.8	94 / 10.3	40 / 6.4	79 / 8.6	23 / 7.2	39 / 8.7
FOMC	89 / 10.1	76 / 9.8	82 / 9.0	77 / 8.9	41 / 9.2	36 / 9.1
Retail	59 / 9.1	77 / 9.8	87 / 9.2	114 / 10.4	26 / 7.6	26 / 7.8
GDP*	95 / 10.0	81 / 9.6	26 / 5.1	145 / 11.2	27 / 7.3	27 / 7.5
FOMC press conf.*	88 / 10.0	111 / 10.9	60 / 8.1	59 / 8.0	45 / 9.5	52 / 10.0
Claims*	124 / 11.0	56 / 9.1	87 / 9.3	81 / 9.1	36 / 9.0	40 / 9.4
ECB*	61 / 9.2	73 / 9.6	88 / 9.0	123 / 10.8	37 / 9.2	42 / 9.6
ECB press conf.*	137 / 11.3	90 / 10.2	38 / 6.5	63 / 8.2	25 / 7.3	32 / 8.0
BoJ*	93 / 10.2	80 / 9.9	45 / 7.3	73 / 8.8	35 / 8.5	37 / 8.8

Table 3: Expected price drift (basis points, end of window) after a +1 standard-deviation surprise, by release and asset (pooled model). *As in Table 2.

Release	EURUSD	USDJPY	Gold	S&P 500	BTC	ETH
CPI	-0.2	+0.1	-0.1	-0.2	-5.3	-1.4
NFP	-0.1	+0.1	+0.1	+0.2	-0.3	-1.5
PPI	0.0	+0.1	-0.2	-0.3	-2.4	-2.1
FOMC	0.0	-0.1	+0.1	+0.1	-0.3	+0.2
Retail	+0.1	-0.1	+0.2	0.0	-1.3	-0.5
GDP*	+0.1	-0.1	-0.1	0.0	+0.9	+0.2
FOMC press conf.*	0.0	0.0	0.0	+0.1	0.0	+0.1
Claims*	0.0	0.0	0.0	0.0	+0.4	+0.6
ECB*	0.0	0.0	0.0	0.0	-0.1	-0.1
ECB press conf.*	0.0	0.0	0.0	0.0	-0.3	-0.3
BoJ*	0.0	0.0	0.0	0.0	-0.1	+0.3

7.3 Track B: public marked-TPP benchmarks

Table 4 compares our models with the published numbers of Chang et al. [4] under the same protocol; our re-run of S2P2 on Taxi (0.524 against the published 0.522) validates the harness. On Retweet, the largest dataset (2.2 million training events), EPT-X improves on the best published model by more than 0.6 nats per event, with a 25-epoch budget imposed by compute and the validation likelihood still rising. It is statistically tied with the best published model on Taobao and StackOverflow, and close on Taxi. There our time term equals S2P2’s (0.733 vs. 0.735 nats per event on the re-run) and the whole difference lies in the mark term. On Amazon, whose inter-event times concentrate in a narrow band (median 0.73, 99th percentile 0.80), the Erlang orders used elsewhere (coefficient of variation $\geq 1/4$) cannot represent the mode; adding atoms of order up to 1024 (coefficient of variation $1/32$), evaluated through the regularised incomplete gamma function, raises EPT-X by 1.1 nats per event on both validation and test, into second place behind S2P2. Neither a deeper (three residual layers) nor a wider (128 units) encoder improved the validation likelihood on any dataset. The renewal channel adds 0.07 nats per event over EPT on Taxi and 0.02 on StackOverflow. The linear MSX model is close to the neural models on Taobao (1.305) and

Retweet, and far behind on Amazon and Taxi, where the dynamics are not self-exciting.

Table 4: Track B, public benchmarks: test log-likelihood per event (higher is better). Published numbers are from Chang et al. [4] (five seeds each); our rows give mean $\pm$ s.d. over seeds (EPT-X: Amazon three; Retweet three; Taxi five; Taobao three; StackOverflow three; other rows one seed), with the EPT-X configuration selected per dataset on validation. Bold: best; underlined: second best; shaded: our models.

	Amazon	Retweet	Taxi	Taobao	StackOverflow
<i>Neural TPPs, published (Chang et al., 2025)</i>					
RMTTPP	−2.136	−7.098	0.346	1.003	−2.480
SAHP	−2.074	−6.708	0.298	1.168	−2.341
THP	−2.096	−6.659	0.372	0.790	−2.338
IntensityFree	0.496	−10.344	0.453	1.318	−2.233
MHP	−2.091	−6.564	0.370	0.636	−2.346
NHP	0.129	<u>−6.348</u>	<u>0.514</u>	1.157	−2.241
AttNHP	0.484	−6.499	0.493	1.259	−2.194
S2P2	0.781	−6.365	0.522	1.304	−2.163
<i>Classical Hawkes, our runs</i>					
ADM4	−2.260	−6.832	−0.315	1.184	−2.473
Conditional law	−	−8.317	−2.922	−63.919	−16.140
Hawkes EM (nonparam.)	−1.002	−6.655	−0.308	1.269	−2.370
<i>Ours</i>					
MSX ($R = 2$)	−1.943	−6.569	−0.214	1.299	−2.292
MSX-auto	−1.927	−6.568	−0.193	1.305	−2.292
EPT	−	−	0.437	1.311	−2.186
EPT-X	<u>0.594</u> ± 0.002	−5.716 ± 0.024	0.503 ± 0.004	<u>1.316</u> ± 0.005	<u>−2.166</u> ± 0.001

7.4 Track C: limit-order-book event streams

Table 5 reports test log-likelihood per event on the five LOBSTER stocks. EPT-X is the best model on every stock, ahead of MSX-auto, EPT and all seven neural baselines; among the stocks on which every baseline has finished, our models win 5 of 5, by 0.23–0.41 nats per event over the best neural baseline. One baseline needs a correction. EasyTPP’s IntensityFree clamps every inter-event time at 10^{-5} s before evaluating its log-normal-mixture density, so all shorter gaps are scored as one common value on which the mixture can place an arbitrarily narrow component. This is not the likelihood of the observed gaps, and it matters most where short gaps are common: 29% and 32% of test gaps on INTC and MSFT lie below 10^{-5} s, against 10–14% on the other three stocks. As shipped, IntensityFree therefore reports the highest numbers on INTC and MSFT (row marked *); re-trained and scored on the unclamped gaps (clamp 10^{-12} s, same 1500-epoch budget) it falls by 1.1 and 0.7 nats per event, below EPT-X and on INTC below MSX. The patch is part of the replication package, and the as-shipped numbers are excluded from the best-model marks. Consistently, neither a denser MSX dictionary (40 instead of 12 rates) nor sharper, history-shifted Erlang atoms in EPT-X change the validation likelihood on INTC by more than 0.05 nats per event: the shipped margin came from the clamp, not from expressiveness. Among the remaining neural baselines, recurrent and attention models with Monte-Carlo compensators (NHP, THP, AttNHP) fall well behind the exact-likelihood phase-type models. Two implementation notes matter for fairness. First, LOBSTER timestamps contain ties, which EasyTPP rejects; all models are trained

and scored on the same files in which tied messages are displaced by $U(0, 100 \text{ ns})$. Second, on our 8 GB GPU the neural baselines run with batch sizes 32–64 instead of 256. Every neural model is trained with a 100-epoch budget and the strongest competitors additionally with longer budgets; for each model the table reports the budget with the best *validation* likelihood, so longer training can only help the baselines. The Hawkes backbone is essential for EPT: without it EPT underperforms plain MSX (superseded run, Appendix). The renewal channel of EPT-X adds 0.08–0.16 nats per event over EPT on every stock.

Table 5: Track C, LOBSTER order books: test log-likelihood per event (higher is better; six event types, 128-event sequences, chronological split; mean $\pm$ s.d. over seeds where several were run). Bold: best; underlined: second best; shaded: our models. *As shipped by EasyTPP, IntensityFree clamps gaps at 10^{-5} s before scoring, which is not the likelihood of the observed data; it is excluded from the ranking, and the IntensityFree row is re-trained and scored on the unclamped gaps.

	AAPL	AMZN	GOOG	INTC	MSFT
<i>Classical Hawkes</i>					
Exp. kernel MLE	1.961	–	−0.311	–	–
Hawkes EM (nonparam.)	3.370	2.243	2.851	5.771	6.258
ADM4	1.961	1.067	1.306	4.787	5.211
Conditional law	−1.018	−17.814	−32.784	0.379	−1.270
<i>Neural TPPs (EasyTPP, our runs)</i>					
RMTTPP	0.243	−0.686	−0.948	2.882	3.285
THP	−0.081	−0.948	−1.075	1.714	1.855
SAHP	1.777	0.462	−0.428	3.915	4.557
NHP	1.374	0.842	0.846	3.874	4.183
AttNHP	1.220	−0.168	0.004	3.497	3.771
S2P2	2.315	0.717	0.990	4.949	5.416
IntensityFree	3.108	2.103	2.726	5.713	6.377
IntensityFree* (as shipped)	3.260	1.921	2.679	6.790	7.070
<i>Ours</i>					
MSX ($R = 2$)	3.326	2.249	2.870	5.812	6.307
MSX-auto	3.361	<u>2.281</u>	<u>2.883</u>	5.814	6.310
EPT	<u>3.380</u> ± 0.008	2.229 ± 0.003	2.806 ± 0.005	<u>6.024</u> ± 0.007	<u>6.530</u> ± 0.005
EPT-X	3.456 ± 0.007	2.333 ± 0.005	2.964 ± 0.004	6.120 ± 0.003	6.643 ± 0.003

7.5 Track D: forecasting the post-release burst

On the 163 releases of 2025–2026, a model fitted on 2022–2024 forecasts the number of δ -crossings in the first 1, 5 and 15 minutes (Table 6). The closed-form MSX forecast improves on same-type climatology at every horizon but is beaten by the event-study OLS and HAR regressions, which are fitted directly to this target; we report this negative result as is. For the *direction* of the post-release move the structural model is informative: the correlation between the forecast and the realised signed drift over the first minute is 0.17, against 0.06 for the best baseline (mean drift by release type and surprise sign).

Table 6: Track D, post-release activity forecasts on the 163 test releases of 2025–2026 (mean over assets; lower is better). Bold: best; underlined: second best; shaded: our model.

Method	RMSE of $\log(1 + N)$			QLIKE of realised variance		
	1 min	5 min	15 min	1 min	5 min	15 min
Climatology	1.440	1.308	1.199	1.263	0.916	0.701
Event-study OLS	1.140	0.925	0.776	2.552	0.580	0.372
HAR	<u>1.217</u>	<u>0.985</u>	<u>0.820</u>	<u>1.467</u>	<u>0.762</u>	<u>0.468</u>
MSX closed form	1.322	1.049	0.955	2.913	1.036	0.518

8 Conclusion

Making every kernel a nonnegative combination of phase-type densities turns Hawkes estimation into a certified convex problem and turns news responses into matrix exponentials. Three findings stand out. First, the “half-life” of news is not a property of the news kernel: echoes lengthen absorption by one to two orders of magnitude, and the correction is available in closed form. Second, falsification matters: our first two specifications produced plausible-looking news effects that the placebo test exposed as artefacts of long kernels, ties and boundary bias. Third, a linear Hawkes backbone and a hyper-Erlang renewal channel make a neural point process exact and the most accurate model on all five order-book streams; benchmark comparisons must, however, check that every model scores the same observed gaps. Limitations: expectations for payrolls, claims, PPI and retail sales are statistical rather than survey-based; the FX feed has a gap in the first half of 2023; neural baselines were trained with reduced batch sizes on an 8 GB GPU; and on the public EasyTPP benchmarks EPT does not reach the state of the art (Section 7).

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A Specification history and the placebo test

We report the specifications that *failed* the built-in falsification test, because the failures are informative and because the final specification was chosen by them, not by the results we hoped for. All superseded outputs are kept in `results/json/superseded/`.

Version 1. News kernels with time scales up to 30 minutes and Erlang order 3 (mean lags up to 90 minutes), a constant baseline per window, time-of-day hats in UTC, and events that could be tied (a trade sweeping several δ levels produced several events with the same millisecond timestamp). The single PLACEBO type absorbed 169 extra events per pseudo-release—comparable to real releases—and crypto residuals had KS statistics up to 0.58. Diagnosis: news kernels longer than the post-release window acted as within-window trend absorbers, and ties (55% of BTC events) violated the orderliness of a simple point process.

Version 2. One event per timestamp (with a size mark), news kernels restricted to 0.1–300s, a per-window *linear* baseline, and New-York-time seasonality. The placebo mass fell to 18.6 and residual fit improved by an order of magnitude, but the placebo remained significant under Wald standard errors. Splitting the placebo by release slot showed placebo mass at every slot, including 14:00 ET on non-FOMC days, which ruled out “unmodelled 08:30 releases” as the whole explanation.

Version 3 (final). Two further changes. (i) Placebo days must avoid sixteen second-tier US releases (PCE, durable goods, trade, housing, JOLTS, sentiment, ...) as well as the modelled ones. (ii) Nonnegative news kernels are estimated with an exposure-weighted ℓ_1 penalty whose level is chosen by the profile likelihood of held-out windows (Table 7); without it, each of the several hundred nonnegative weights has positive expected value under the null, so release and placebo masses are equally inflated (boundary bias). The held-out likelihood peaks at $\ell_1 = 0.8$, where release and placebo masses separate.

Table 7: Selection of the news-kernel penalty on held-out 2024 windows (profile log-likelihood relative to the best level).

ℓ_1 level	held-out LL (rel.)	mean release mass	mean placebo mass
0.2	-93.7	10.79	6.23
0.4	-27.1	5.20	2.89
0.8	0.0	1.81	1.03
1.6	-20.5	0.44	0.37

B EPT ablations

Hawkes backbone. Without the explicit linear Hawkes state, EPT reached 2.84 nats per event on LOBSTER AAPL (three seeds, 100 epochs), below the classical MSX estimator; the network had to rediscover mark-specific linear excitation through its gated jumps and did not within the epoch budget. With the backbone—a residual path that makes the model exactly MSX when the neural read-outs vanish—EPT exceeds every neural baseline on all five stocks and MSX on three of them (Table 5).

Renewal channel and encoder. Adding the hyper-Erlang renewal channel (EPT-X) improves EPT on every LOBSTER stock and on every public dataset where both were run. On validation, sharper Erlang atoms (orders up to 1024) help only on Amazon, whose gaps concentrate in a narrow band, and neither a three-layer residual encoder nor a 128-unit encoder improves any dataset (Section 7).

C Track A: all metrics

Table 8: Track A, all metrics (mean $\pm$ s.d. over five seeds): relative branching-matrix error, spectral-radius error, kernel L^1 error, held-out deficit dLL (10^{-3} nats/event) and edge AUC.

Scenario	Method	G err.	ρ err.	kernel L^1	dLL (10^{-3})	AUC
S1-exp	MSX-auto (ours)	0.054 ± 0.032	0.020 ± 0.018	0.093	0.24 ± 0.12	–
	MSX (ours)	0.056 ± 0.032	0.019 ± 0.018	0.092	0.23 ± 0.12	–
	MSX-R4 (ours, fixed Erlang-4)	0.056 ± 0.035	0.023 ± 0.018	0.105	0.28 ± 0.15	–
	MSX-exp (ablation: no Erlang)	0.055 ± 0.025	0.026 ± 0.015	0.157	0.91 ± 0.13	–
	ExpKern(MLE, best decay)	0.018 ± 0.007	0.005 ± 0.003	0.018	0.05 ± 0.05	–
	SumExpKern(MLE, AGD)	0.065 ± 0.022	0.030 ± 0.013	0.165	0.92 ± 0.13	–
	ADM4	0.018 ± 0.007	0.005 ± 0.003	0.018	0.05 ± 0.05	–
	EM(nonparam)	0.035 ± 0.015	0.017 ± 0.007	0.173	1.02 ± 0.07	–
	NPHC(cumulants)	0.247 ± 0.087	0.032 ± 0.024	–	–	–
	ConditionalLaw	0.120 ± 0.054	0.017 ± 0.016	0.193	2.84 ± 1.03	–
S2-powerlaw	MSX-auto (ours)	0.064 ± 0.029	0.037 ± 0.029	0.108	0.33 ± 0.13	–
	MSX (ours)	0.057 ± 0.021	0.015 ± 0.010	0.114	0.32 ± 0.12	–
	MSX-R4 (ours, fixed Erlang-4)	0.062 ± 0.014	0.016 ± 0.006	0.131	0.45 ± 0.10	–
	MSX-exp (ablation: no Erlang)	0.054 ± 0.022	0.014 ± 0.010	0.097	0.19 ± 0.11	–
	ExpKern(MLE, best decay)	0.299 ± 0.011	0.219 ± 0.007	0.465	26.05 ± 0.50	–
	SumExpKern(MLE, AGD)	0.057 ± 0.012	0.016 ± 0.007	0.094	0.18 ± 0.12	–
	ADM4	0.299 ± 0.011	0.219 ± 0.007	0.465	26.05 ± 0.50	–
	EM(nonparam)	0.057 ± 0.014	0.037 ± 0.008	0.188	1.73 ± 0.21	–
	NPHC(cumulants)	0.153 ± 0.154	0.041 ± 0.047	–	–	–
	ConditionalLaw	0.102 ± 0.067	0.029 ± 0.019	0.291	4.91 ± 0.59	–
S3-hump	MSX-auto (ours)	0.032 ± 0.016	0.016 ± 0.007	0.080	0.30 ± 0.13	–
	MSX (ours)	0.069 ± 0.009	0.037 ± 0.004	0.347	5.90 ± 0.36	–
	MSX-R4 (ours, fixed Erlang-4)	0.038 ± 0.010	0.017 ± 0.005	0.074	0.31 ± 0.15	–
	MSX-exp (ablation: no Erlang)	0.034 ± 0.009	0.015 ± 0.004	0.416	11.51 ± 0.46	–
	ExpKern(MLE, best decay)	0.166 ± 0.008	0.053 ± 0.004	0.801	101.47 ± 1.81	–
	SumExpKern(MLE, AGD)	0.038 ± 0.011	0.018 ± 0.005	0.419	11.51 ± 0.46	–
	ADM4	0.166 ± 0.008	0.053 ± 0.004	0.801	101.47 ± 1.81	–
	EM(nonparam)	0.024 ± 0.011	0.010 ± 0.004	0.187	2.24 ± 0.12	–
	NPHC(cumulants)	0.240 ± 0.096	0.045 ± 0.048	–	–	–
	ConditionalLaw	0.149 ± 0.062	0.017 ± 0.010	0.319	10.66 ± 2.13	–
S4-multiscale	MSX-auto (ours)	0.084 ± 0.014	0.016 ± 0.006	0.128	0.32 ± 0.08	1.000
	MSX (ours)	0.093 ± 0.015	0.021 ± 0.005	0.173	0.54 ± 0.12	1.000
	MSX-R4 (ours, fixed Erlang-4)	0.101 ± 0.019	0.027 ± 0.007	0.207	0.70 ± 0.11	1.000
	MSX-exp (ablation: no Erlang)	0.084 ± 0.014	0.016 ± 0.006	0.128	0.32 ± 0.08	1.000
	ExpKern(MLE, best decay)	0.173 ± 0.041	0.095 ± 0.025	1.635	711.28 ± 53.09	1.000
	ADM4	0.399 ± 0.003	0.255 ± 0.001	0.410	3.13 ± 0.25	1.000
	EM(nonparam)	0.153 ± 0.012	0.045 ± 0.010	0.358	4.55 ± 0.29	1.000
	NPHC(cumulants)	1.030 ± 0.304	0.211 ± 0.186	–	–	0.588
	ConditionalLaw	0.464 ± 0.286	0.139 ± 0.050	1.059	114.94 ± 1.20	0.831
S5-network10	MSX-auto (ours)	0.045 ± 0.004	0.015 ± 0.005	0.079	0.74 ± 0.10	1.000
	MSX (ours)	0.111 ± 0.013	0.040 ± 0.013	0.180	1.19 ± 0.07	1.000
	MSX-R4 (ours, fixed Erlang-4)	0.122 ± 0.008	0.051 ± 0.013	0.218	1.63 ± 0.09	1.000
	MSX-exp (ablation: no Erlang)	0.099 ± 0.014	0.030 ± 0.014	0.145	0.82 ± 0.08	1.000
	ExpKern(MLE, best decay)	0.337 ± 0.153	0.103 ± 0.032	1.302	426.80 ± 368.57	0.977
	ADM4	0.167 ± 0.003	0.096 ± 0.003	0.319	26.51 ± 0.29	1.000
	EM(nonparam)	0.085 ± 0.008	0.045 ± 0.008	0.280	5.23 ± 0.25	1.000
	NPHC(cumulants)	1.348 ± 0.477	0.520 ± 0.372	–	–	0.672
	ConditionalLaw	0.461 ± 0.047	0.037 ± 0.026	0.641	78.64 ± 4.79	0.950